

Learning-based optimisation framework for flexibility aggregators in ancillary service markets

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Abstract

This paper presents a learning-based optimisation framework for flexibility aggregators participating in the European manual frequency restoration reserve (mFRR) market. Accurately modelling prosumer responsiveness is critical for developing an aggregator's optimal bidding strategy, yet it is computationally challenging due to its nonlinear and uncertain nature. To address this, we propose a novel and tight linear programming (LP) reformulation for rectified linear unit (ReLU) deep neural networks (DNNs), enabling the learning of prosumer behaviour and its tractable integration within an aggregator's portfolio optimisation problems. This formulation leverages the convex nature of non-negatively weighted ReLU DNNs, referred to as convexified ReLU DNNs, to ensure tractable and tight embedding within minimisation problems. The proposed method is benchmarked against state-of-the-art ReLU DNN reformulations within a flexibility aggregator's profit maximisation problem. Simulation results show that the convexified ReLU DNN reformulation consistently achieves near-optimal solution quality with significantly lower computational burden, outperforming existing ReLU DNN reformulations in terms of accuracy, tractability, and tightness across varying ReLU DNN architectures. These findings highlight the potential of the proposed approach for real-time, scalable applications in power system optimisation, particularly where data-driven models are needed to represent complex, nonconvex behaviours such as prosumer flexibility.

Keywords: constraint learning, neural networks, electricity market, flexibility aggregator, demand response

List of Abbreviations

ACER	Agency for Cooperation of Energy Regulators
aFRR	Automatic frequency restoration reserve
BTM	Behind-the-meter
cvxd	convexified
DER	Distributed energy resources
DNN	Deep neural network
DR	Demand response
DSF	Demand side flexibility
FBBT	Feasibility-based bound tightening
HEMS	Home energy management system
LP	Linear program
mFRR	Manual frequency restoration reserve
MIP	Mixed integer program
MSE	Mean square error
OBBT	Optimisation-based bound tightening
OCL	Optimisation with constraint learning
PCAR	Penalised convex area relaxation
PCTAR	Penalised convex triangle area relaxation
PWL	Piecewise linearisation
ReLU	Rectified linear unit
RMSE	Root mean square error
SOTA	State-of-the-art
UC	Unconstrained

Nomenclature

Indices and Sets

$\mathbb{N}$	Set of natural numbers
$\mathbb{R}$	Set of real numbers
$\mathbb{R}_+$	Set of non-negative real numbers
$\mathcal{L}$	Index of ReLU DNN layers, $\{1, \dots, L + 1\}$
$\mathcal{T}$	Set of time steps, $\{1, \dots, T\}$

Parameters

α_l	Penalty coefficients for hidden layer $l \in \mathcal{L}$, $\alpha_l \in \mathbb{R}^{n_l}$
λ^B	mFRR activation prices in DKK/MWh, $\lambda^U, \lambda^D \in \mathbb{R}^T$
λ^U, λ^D	mFRR capacity up- and down-regulation prices in DKK/MWh, $\lambda^U, \lambda^D \in \mathbb{R}^T$
$\bar{x}^u, \bar{x}^d$	Maximum obtainable up- and down-regulation DSF in MWh, $\bar{x}^u, \bar{x}^d \in \mathbb{R}_+^T$
$\mathbf{A}^{u,\downarrow}, \mathbf{A}^{d,\downarrow}$	Down-regulation activated DSF cross-temporal matrices, $\mathbf{A}^{u,\downarrow}, \mathbf{A}^{d,\downarrow} \in [0, 1]^{T \times T}$

$\mathbf{A}^{\text{u},\uparrow}, \mathbf{A}^{\text{d},\uparrow}$	Up-regulation activated DSF cross-temporal matrices, $\mathbf{A}^{\text{u},\uparrow}, \mathbf{A}^{\text{d},\uparrow} \in [0, 1]^{T \times T}$
$\mathbf{b}_l$	Bias matrix of hidden layer $l \in \mathcal{L}$, $\mathbf{b}_l \in \mathbb{R}^{n_l}$
$\mathbf{q}^{\text{d}}, \mathbf{r}^{\text{d}}$	Down-regulation prosumer responsiveness shaping parameters, $\mathbf{q}^{\text{d}}, \mathbf{r}^{\text{d}} \in \mathbb{R}_+^T$
$\mathbf{q}^{\text{u}}, \mathbf{r}^{\text{u}}$	Up-regulation prosumer responsiveness shaping parameters, $\mathbf{q}^{\text{u}}, \mathbf{r}^{\text{u}} \in \mathbb{R}_+^T$
$\mathbf{W}_l$	Weight matrix of hidden layer $l \in \mathcal{L}$, $\mathbf{W}_l \in \mathbb{R}^{n_{l-1} \times n_l}$
L	Number of hidden ReLu layers, $L \in \mathbb{N}$
LB, UB	Bounds for PCTAR reformulation, $LB, UB \in \mathbb{R}$
N_D	Number of training data points, $N_D \in \mathbb{N}$
n_l	Number of neurons in hidden layer $l \in \mathcal{L}$, $n_l \in \mathbb{N}$
N_P	Number of pieces for PWL, $N_P \in \mathbb{N}$
T	Optimisation horizon, $T \in \mathbb{N}$

Variables

$\lambda^{\text{I},\text{u}}, \lambda^{\text{I},\text{d}}$	Per-unit price paid for reserved up- and down-DSF in DKK/MWh, $\lambda^{\text{I}} \in \mathbb{R}^T$
$\lambda^{\text{P},\text{u}}, \lambda^{\text{P},\text{d}}$	Total cost paid for up- and down-DSF in DKK, $\lambda^{\text{P},\text{u}}, \lambda^{\text{P},\text{d}} \in \mathbb{R}_+^T$
λ^{P}	Total cost paid by aggregator to prosumer in DKK, $\lambda^{\text{P}} \in \mathbb{R}_+^T$
$\tilde{\mathbf{x}}^{\text{u}}, \tilde{\mathbf{x}}^{\text{d}}$	Updated maximum up- and down-regulation DSF in MWh, $\tilde{\mathbf{x}}^{\text{u}}, \tilde{\mathbf{x}}^{\text{d}} \in \mathbb{R}_+^T$
$\mathbf{h}_l$	Output vector of hidden layer $l \in \mathcal{L}$, $\mathbf{h}_l \in \mathbb{R}^{n_l}$
$\mathbf{x}^{\text{u}}, \mathbf{x}^{\text{d}}$	Reserved up- and down-DSF for mFRR capacity market in MWh, $\mathbf{x}^{\text{u}}, \mathbf{x}^{\text{d}} \in \mathbb{R}_+^T$

1. Introduction

1.1. Motivation and related work

To sustain the rapid growth of renewable energy sources, the European Union Agency for the Cooperation of Energy Regulators (ACER) projects that by 2030, the flexibility capacity of the grid will need to double from its current capacity [1]. As conventional generators, once the main source of flexibility on the supply side, are being replaced by renewable sources, demand response (DR) has become a critical mechanism to achieve the necessary flexibility of the network [2]. DR mechanisms enable end consumers, including prosumers (those who consume and produce electricity), to adjust their electricity production or consumption in response to time-varying incentives, whether financial or otherwise [3].

An extremely popular DR mechanism is the indirect load control of aggregated prosumers. Here, aggregators of behind-the-meter (BTM) flexibility leverage the growing penetration of distributed energy resources (DER), such as rooftop solar panels, electric vehicles, and batteries along with notable advances in home energy management systems (HEMS) to coordinate prosumers in a grid-aware manner. This coordination enables the

acquisition of demand-side flexibility (DSF), which can be offered to the energy and ancillary services markets. This coordination yields significant benefits: consumers receive financial incentives for shifting their production or consumption, while system operators gain access to cost-effective and fast-acting DSF. Consequently, state-of-the-art (SOTA) research focuses heavily on aggregator business models and optimal bidding strategies [4]. For aggregators to operate profitably, accurately modelling prosumers' price responsiveness is critical. It is essential for these models to accurately assess the DSF provided by the prosumers because the system operators apply financial penalties and limit future market participation if promised services are not delivered when needed [5]. In addition, these models enable aggregators to assess various business strategies and engage confidently in the market through optimal bidding. As noted by the authors in [6], prosumers' marginal utility functions or responsiveness functions are often nonlinear, non-smooth, and heavily influenced by uncertainties in electricity prices, load patterns, and weather conditions, making them difficult to model explicitly. Additionally, non-ideal or irrational behaviour of prosumers may further exacerbate the above challenge [7]. To effectively address these modelling challenges and encapsulate the complex, uncertain, and potentially irrational behaviours exhibited by prosumers, the optimisation with constraint learning (OCL) framework can offer a robust and systematic approach.

An OCL framework employs some machine learning-based surrogate models to approximate objective and constraint functions that are difficult to explicitly formulate within optimisation models while ensuring that critical constraints are satisfied [8]. This framework is gaining prominence in modern power system research due to its ability to learn complex, uncertain, or nonlinear interactions in optimisation problems, such as optimal power flow [9] and unit commitment [10]. Furthermore, OCL can dynamically model changes in the stochastic behaviour of parameters with respect to decision variables, as demonstrated in [11], and more accurately capture the irrational human behaviour that often influences optimal decision-making [12]. These features position an OCL framework as a promising tool for modelling prosumers' price responsiveness.

The authors in [8] conducted a comprehensive review and identified ReLU-based deep neural networks (DNNs) as one of the most commonly employed machine learning methodologies within the OCL framework. This preference can be attributed to their continuous piecewise linear nature, which enables them to capture a wide range of complex relationships. ReLU DNNs are highly versatile, making them suitable for regression and classification tasks [13, 14, 15]. Moreover, tight mixed-integer programming (MIP) reformulations exist for embedding ReLU DNNs into optimisation models, both in the literature and as Python packages [16, 17]. These include the Big M formulation for max functions, augmented with feasibility-based bound tightening (FBBT) or optimality-based bound tightening (OBBT) algorithms to refine the bounds of the ReLU neuron outputs. In FBBT, bounds are tightened by adding cuts to the feasibility space without removing any feasible solutions, whereas in OBBT, bounds are refined by solving a series of optimisation problems. Both methods significantly reduce the search space and improve the computational efficiency of the Big M formulation [15]. However, MIP reformulations introduce a binary variable for each ReLU neuron, which can substantially increase computational complexity. To maintain tractability, most researchers opt for extremely shallow ReLU DNNs when embedding them into optimisation models [8].

The authors in [18] encountered the challenge of intractability while using ReLU DNN to learn and model battery degradation, where they used the Big M reformulation of ReLU DNN to schedule multiple batteries. To solve the intractability issue, the same

authors in [19] proposed two linear programming (LP) reformulations named penalised convex area relaxation (PCAR) and penalised convex triangle area reformulation (PCTAR), which are discussed in more detail in Section 3.2. By implementing these modifications, the authors observed an improvement in the tractability of the scheduling problem. However, they pointed out that while PCAR lacked accuracy, PCTAR was accurate and manageable. Notably, both PCAR and PCTAR are penalty-driven methods, relying on model-specific tuning and showing sensitivity to the optimisation formulation, ReLU DNN architecture, and hyperparameter choices, as demonstrated in the sensitivity analysis in Section 4.5. This makes them less robust across different applications.

Despite growing interest in using ReLU DNNs as surrogate models within the OCL framework, two key research gaps remain. First, existing LP-based reformulations, those proposed in [19], compromise guaranteed tightness in favour of tractability. Second, to the best of our knowledge, no previous studies have applied the OCL framework to model prosumer responsiveness within a profit maximisation problem of a BTM flexibility aggregator. Given the complexity and scale of BTM aggregation, this represents a critical gap in both the methodological development of OCL and its practical application to DR optimisation problems.

1.2. Objectives and contributions

To address the main identified gaps, we develop a profit maximisation framework for an aggregator of BTM flexibility participating in the European manual frequency restoration reserve (mFRR) capacity market. Central to this formulation is the modelling of the prosumer responsiveness function, which is inherently nonlinear and nonconvex. To maintain tractability, we approximate this function using a ReLU DNN and embed it within the aggregator’s optimisation problem via our proposed LP reformulation. The key contributions of this work are as follows:

- Proposal of a method to “convexify” ReLU DNNs, enabling their integration as tight LP reformulations within optimisation problems.
- Development of an aggregator’s profit maximisation model incorporating a nonlinear, nonconvex prosumer responsiveness function. The prosumer responsiveness function is a modified version of our previous published work [20].
- Application of the OCL framework by learning the prosumer responsiveness function using a convexified ReLU DNN (cvxd ReLU DNN) and embedding it as an LP reformulation within the aggregator’s profit maximisation problem.
- Comprehensive comparison of the proposed reformulation against SOTA ReLU DNN reformulations and the conventional piecewise linearisation (PWL), evaluated across a range of optimisation parameters and ReLU DNN architectures.

We note that the proposed ReLU DNN reformulation is only applicable to a function or decision variable in a minimisation problem, which we further elaborate in Section 3.1. Despite this caveat, it offers a strong potential for broader use in modern power system applications, including, but not limited to, minimisation of nonlinear generator cost functions and CO₂ emissions.

It is important to emphasise that the primary objective of this paper is to demonstrate the properties and effectiveness of the proposed ReLU DNN reformulation and to benchmark it against the SOTA methods. The optimisation model and case study presented here

serve to support this objective rather than being the main focus. While the mFRR capacity market is used as the case study, the proposed formulation is generalisable to other market contexts involving BTM aggregator participation.

The remainder of the paper is structured as follows: Section 2 presents the mathematical formulation of the OCL case study. Section 3 details the surrogate models used to represent prosumer behaviour. Section 4 outlines the simulation setup, presents the results, and provides a comprehensive analysis. Finally, Section 5 concludes the paper and outlines directions for future work.

2. Problem statement

In this section, we describe the background information, assumptions, and mathematical models used to develop a profit maximisation problem for a BTM flexibility aggregator.

2.1. Background and assumptions

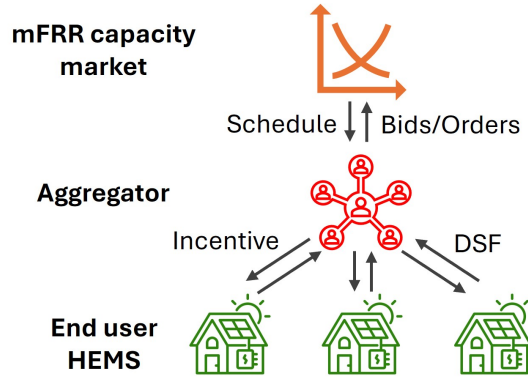


Figure 1: BTM flexibility aggregator business model

Figure 1 illustrates the aggregator’s business model used in this case study. Although in this study, we applied this business model specifically to the mFRR capacity market, it is adaptable to other markets. The aggregator sends an incentive price signal to the HEMS of multiple prosumers. Subsequently, prosumers react by providing a certain amount of flexibility. The flexibility purchased by the aggregator can then be used to participate in various electricity markets and services. Because each of these markets and services varies significantly depending on the region, we limit the scope of this case study to the European mFRR capacity market operating in Denmark, which is operated by the Danish transmission system operator Energinet.

The mFRR capacity market is a pay-as-bid market, used to procure tertiary or replacement reserves in both directions up and down to balance demand and supply in case of emergencies. Up-regulation involves exporting energy to the grid or reducing self-consumption, whereas down-regulation involves importing energy from the grid or increasing consumption. This market is unsymmetrical in nature; the market participants can choose to provide up-regulation, down-regulation or both depending on their business strategy.

These markets operate a day before delivery and ensure that reserves are available every hour the next day. These reserves are activated in the event of unexpected emergencies. As a capacity market, all resources acquired through the market process are paid at mFRR capacity market prices, regardless of whether the reserve is activated. Hence,

market participants whose bids are accepted must ensure that the accepted reserve is available; otherwise, they are fiscally liable.

Actual activations are coordinated via the mFRR activation market, a separate pay-as-bid market held 45 minutes prior to the delivery interval. Energinet employs a one-price imbalance settlement system, whereby a single imbalance price is used to settle both up- and down-regulation at a given interval. According to market regulations, any entity whose bid is accepted in the mFRR capacity market must offer at least the accepted volume in the activation market. Since the activation market operates closer to real-time, aggregators may choose to offer additional volume beyond their accepted capacity bid if they anticipate greater flexibility availability as the delivery interval approaches. For further details on the ancillary service markets and regulatory guidelines in Denmark, the readers are referred to [5].

Based on the described aggregator and market setting, we make the following assumptions to formulate the aggregator’s profit maximisation problem.

- The aggregator acts as a price-taking entity, and hence their bids are always fully accepted. Furthermore, market prices for both the capacity and activation markets, as well as mFRR activation signals, are assumed to be known deterministically.
- The aggregator deterministically knows the maximum obtainable flexibility from their prosumer cluster at each interval.
- The rebound effect on prosumers’ flexibility is considered. Specifically, up-regulation flexibility activation at a given interval reduces the maximum obtainable up-regulation flexibility and increases the maximum obtainable down-regulation flexibility in subsequent intervals. Conversely, down-regulation flexibility activation decreases future down-regulation potential while increasing up-regulation availability. For example, discharging a prosumer’s battery in the current interval necessitates recharging in later intervals, thereby increasing future consumption and reducing export capability. This rebound effect is captured through cross-temporal weights, which are assumed to be known deterministically.
- Given the deterministic nature of the problem, the aggregator can precisely determine the optimal volume to bid on the mFRR capacity market. Therefore, it only bids the volume accepted in the capacity market and does not offer any additional flexibility in the activation market.

Based on these assumptions, we formulate a deterministic profit-maximising model for an aggregator participating in the mFRR markets. While real-world aggregator operations are inherently uncertain and require robust or stochastic formulations to capture price, activation, and flexibility availability uncertainties, such models introduce additional computational complexity. Since the primary aim of this study is to evaluate the performance of our proposed ReLU DNN formulation and benchmark it against the existing SOTA, we construct a deterministic optimisation framework to ensure computational tractability for all the surrogate models described in Section 3. This simplification enables a fair and comprehensive comparison of all implemented surrogate models for prosumer responsiveness within the optimisation framework. For a more in-depth exploration of complex and uncertainty-aware aggregator business models and market participation strategies, the reader is referred to [4] and [21].

We also note that our formulation is readily extendable to incorporate uncertainty in prices, mFRR activations, and flexibility volumes for both up- and down-regulation, using approaches such as scenario-based modelling, chance constraints, or other stochastic methods. Furthermore, it can be applied directly with minimal or no modifications to other European ancillary services markets, such as the automatic frequency restoration reserve (aFRR).

2.2. Mathematical models

Based on the background and assumptions stated in Section 2.1, we develop a profit maximisation problem for a BTM flexibility aggregator participating in the European mFRR capacity market.

Let $\mathcal{T} = \{1, \dots, T\}$ denote the set of time intervals for which the problem is solved with the time horizon $T \in \mathbb{N}$. Let $\boldsymbol{\lambda}^U, \boldsymbol{\lambda}^D \in \mathbb{R}^T$ represent the up- and down-regulation price profiles of the mFRR capacity market in DKK/MWh, $\boldsymbol{\lambda}^B \in \mathbb{R}^T$ represent the imbalance price profiles of the mFRR activation market in DKK/MWh, and $\boldsymbol{a}^\uparrow, \boldsymbol{a}^\downarrow \in [0, 1]$ denote the mFRR up- and down-regulation activation signal from Energinet.

Let $\boldsymbol{x}^u, \boldsymbol{x}^d \in \mathbb{R}_+^T$ denote the reserved up- and down-regulation flexibility for the mFRR capacity market in MWh, $\boldsymbol{x}^\uparrow, \boldsymbol{x}^\downarrow \in \mathbb{R}_+^T$ denote the activated up- and down-regulation flexibility for the mFRR market in MWh, and $\bar{\boldsymbol{x}}^u, \bar{\boldsymbol{x}}^d \in \mathbb{R}_+^T$ correspond to the maximum obtainable up- and down-flexibility from a prosumer group in MWh. Due to the rebound effect, the maximum obtainable up- and down-flexibility are updated to $\tilde{\boldsymbol{x}}^u, \tilde{\boldsymbol{x}}^d \in \mathbb{R}_+^T$ in MWh respectively. Finally, let $\boldsymbol{\lambda}^{I,u}, \boldsymbol{\lambda}^{I,d} \in \mathbb{R}_+^T$ be the incentive offered to the prosumer group in DKK/MWh to provide up- and down-flexibility respectively.

To maintain conciseness, we describe the prosumer responsiveness function from the perspective of up-regulation; the formulation applies analogously to down-regulation. We define the prosumer responsiveness function based on our previous work in [20], as given in Eq. (1a). This function models the relationship using a saturation curve, as prosumers are typically not willing to reduce consumption at low prices, with their up-regulation flexibility gradually saturating to the updated maximum achievable up-flexibility $\tilde{x}_t^u$, at very high prices. Notably, this approach modifies the methodology proposed in [20] by substituting $\tilde{\boldsymbol{x}}^u$ for $\bar{\boldsymbol{x}}^u$. This substitution provides a more realistic representation of the prosumer responsiveness, as the rebound effect appreciates the true value of potential future flexibility. For example, a battery discharged in the current interval requires a greater incentive to discharge in the future intervals. This is because its future flexibility is reduced, making it a more valuable resource.

We can see that the prosumers responsiveness function depends on the decision vectors $\boldsymbol{x}^i, \tilde{\boldsymbol{x}}^i$, and the saturation curve shaping variables $q_t^i, r_t^i \in \mathbb{R}_+$ where $i \in \{u, d\}$ represents up- and down-regulation and $t \in \mathcal{T}$. Parameters q_t^i, r_t^i are used to capture exogenous factors relevant to this relationship for both up- and down-regulation $i \in \{u, d\}$ and at time $t \in \mathcal{T}$. Notably, these shaping parameters are expected to differ between up- and down-regulation, reflecting temporal variations in prosumer preferences. For example, during peak-load periods, prosumers often prefer increasing consumption (i.e., reducing export) rather than curtailing it, resulting in a higher valuation of up-regulation flexibility. Conversely, for prosumers with specific DER configurations, such as solar PV without storage, down-regulation flexibility may be valued more highly than up-regulation during mid-day periods.

To avoid bilinear terms within the objective function, we rearrange Eq. (1a) into Eq. (1c), which evaluates the total purchase cost incurred by the aggregator to obtain the reserved

up- and down-regulation flexibility. This cost is denoted by $\lambda_t^{\text{P,u}}, \lambda_t^{\text{P,d}} \in \mathbb{R}_+$ at time $t \in \mathcal{T}$ expressed in DKK. Please note that max function in Eq. (1c) ensures that the incentive paid to prosumers is non-negative. Clearly, this relationship is complex and introduces computational challenges when applied to optimisation problems.

$$x_t^i = \frac{\tilde{x}_t^i}{1 + e^{(q_t^i - r_t^i \lambda_t^{\text{I},i})}} \quad \forall t \in \mathcal{T}, i \in \{\text{u,d}\} \quad (1a)$$

$$\Rightarrow \lambda_t^{\text{I},i} = \frac{1}{r_t^i} \left[q_t^i - \ln \left(\frac{\tilde{x}_t^i}{x_t^i} - 1 \right) \right] \quad \forall t \in \mathcal{T}, i \in \{\text{u,d}\} \quad (1b)$$

$$\therefore \lambda_t^{\text{P},i} = \max \left\{ \frac{x_t^i}{r_t^i} \left[q_t^i - \ln \left(\frac{\tilde{x}_t^i}{x_t^i} - 1 \right) \right], 0 \right\} \quad \forall t \in \mathcal{T}, i \in \{\text{u,d}\} \quad (1c)$$

Thus, we can define the BTM flexibility aggregator's profit maximisation problem as shown in Eqs. (2a) – (2f):

$$\max_{\mathbf{x}, \tilde{\mathbf{x}}, \boldsymbol{\lambda}^{\text{P}}} \sum_{t \in \mathcal{T}} \left[\lambda_t^{\text{U}} x_t^{\text{u}} + \lambda_t^{\text{D}} x_t^{\text{d}} + \lambda_t^{\text{B}} (x_t^{\uparrow} - x_t^{\downarrow}) - \lambda_t^{\text{P}} \right] \quad (2a)$$

$$\text{s.t.} \quad \tilde{x}_t^{\text{u}} = \bar{x}_t^{\text{u}} - \sum_{j \in \mathcal{T}} A_{tj}^{\text{u},\uparrow} x_t^{\uparrow} + \sum_{j \in \mathcal{T}} A_{tj}^{\text{u},\downarrow} x_t^{\downarrow} \quad \forall t \in \mathcal{T} \quad (2b)$$

$$\tilde{x}_t^{\text{d}} = \bar{x}_t^{\text{d}} - \sum_{j \in \mathcal{T}} A_{tj}^{\text{d},\downarrow} x_t^{\downarrow} + \sum_{j \in \mathcal{T}} A_{tj}^{\text{d},\uparrow} x_t^{\uparrow} \quad \forall t \in \mathcal{T} \quad (2c)$$

$$x_t^{\uparrow} = a_t^{\uparrow} x_t^{\text{u}}, x_t^{\downarrow} = a_t^{\downarrow} x_t^{\text{d}} \quad \forall t \in \mathcal{T} \quad (2d)$$

$$x_t^i \leq \tilde{x}_t^i \quad \forall t \in \mathcal{T}, i \in \{\text{u, d}\} \quad (2e)$$

$$\lambda_t^{\text{P}} = \lambda_t^{\text{P,u}} + \lambda_t^{\text{P,d}} \quad \forall t \in \mathcal{T} \quad (2f)$$

Equation (2a) sets the objective of the optimisation problem as maximisation of the net profit of the aggregator. Equation (2d) specifies the activated mFRR up- and down-regulation flexibility. Equation (2e) ensures that the reserved up- and down-regulation flexibility is within the corresponding updated maximum available flexibility. Equations (2b) and (2c) account for the rebound effect using the cross-temporal matrices $\mathbf{A}^{\text{u},\uparrow}, \mathbf{A}^{\text{u},\downarrow}, \mathbf{A}^{\text{d},\uparrow}, \mathbf{A}^{\text{d},\downarrow} \in [0, 1]^{T \times T}$, which are lower triangular matrices with diagonal elements equal to zero. If up-regulation flexibility is activated at some interval t , the available up-regulation flexibility in subsequent intervals is reduced, while the potential for down-regulation flexibility increases. In contrast, activation of down-regulation flexibility at time t decreases future down-regulation availability and enhances the up-regulation potential. Equation (2f) corresponds to the total purchase cost for both up- and down-regulation flexibility obtained from prosumers.

As defined in Eq. (1c), $\forall i \in \{\text{u,d}\}$, $\lambda_t^{\text{P},i}$ is both nonlinear and nonconvex presenting computational challenges within the profit maximisation problem. Hence, we replace this cost function with a tractable surrogate model $\tilde{f}(x_t^i, \tilde{x}_t^i, q_t^i, r_t^i)$, $\forall i \in \{\text{u,d}\}$, which approximates the original purchase cost function. If $\tilde{f}(\cdot)$ is based on an ML technique, the optimisation model falls under the OCL framework. The surrogate models adopted in this study are further elaborated in Section 3.

3. Surrogate models for DSF purchase cost $\lambda^{\text{P},i}$

In this section, we describe the various surrogate models of $\lambda^{\text{P},i}$ used in this study. Here, we consider PWL [22] and ReLU DNN-based approximations to replace the nonlinear function in Eq. (1c). The reformulations of the ReLU DNN considered include:

1. The standard MIP reformulation implemented via Gurobi machine learning (Gurobi ML) Python package [16].
2. The LP reformulations presented in [19], namely PCAR and PCTAR.
3. Our proposed reformulation presented in Theorem 3.1.

PWL methods are ubiquitous for approximating nonlinear and nonconvex functions in the optimisation literature. However, they suffer the “curse of dimensionality” with both increasing number of inputs and increasing number of pieces [15], making them computationally inefficient for complex relationships. The responsiveness of prosumers depends on a multitude of factors with most likely no closed-form expressions, making PWL impractical for this application. Nonetheless, we treat PWL as a baseline in this study because of its ubiquity.

Evidently, the purchase cost specified in Eq. (1c) is a multivariate function. Applying the PWL technique to linearise this function will be computationally challenging for even a small number of pieces N_P , due to the input size and time horizon T . Hence, we simplify Eq. (1c) to a two-dimensional function by assuming that the exogenous factors are deterministically known; that is, $q_t^i, r_t^i, \forall i \text{ in } \{u, d\}$ are known beforehand. We then employ the work of the authors in [22] to implement the PWL approximation.

In contrast, ReLU DNNs can easily learn multivariate functions. Hence, we treat the nonlinear function as a four-dimensional function to train the ReLU DNNs. In the rest of this section, we summarise the SOTA and our proposed ReLU DNN reformulations, and the ReLU DNN training strategy employed in this study.

3.1. Proposed ReLU DNN reformulation

In this section, we discuss the mathematical background, intuition, and key properties of the proposed ReLU DNN reformulation.

Definition 1. For any vector $\mathbf{z} \in \mathbb{R}^a$ with any $a \in \mathbb{N}$, we can define the element-wise ReLU function $\sigma : \mathbb{R}^a \rightarrow \mathbb{R}_+^a$, as shown in (3)

$$\sigma(\mathbf{z}) = \begin{bmatrix} \max(z_1, 0) \\ \max(z_2, 0) \\ \vdots \\ \max(z_a, 0) \end{bmatrix} \quad (3)$$

Definition 2. Consider a fully connected ReLU DNN $f : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^{n_{L+1}}$, where $n_0 \in \mathbb{N}$ is the input size and $L \in \mathbb{N}$ is the number of ReLU hidden layers. Let $n_l \in \mathbb{N}$ be the number of output neurons with weights $\mathbf{W}_l \in \mathbb{R}^{n_{l-1} \times n_l}$ and biases $\mathbf{b}_l \in \mathbb{R}^{n_l}$ for each layer $l \in \{1, 2, \dots, L, L+1\}$, where $l = L+1$ is the output layer with n_{L+1} outputs. Given an input vector $\mathbf{z} \in \mathbb{R}^{n_0}$ such that $\mathbf{h}_0 = \mathbf{z}$, the output of the hidden layers $\mathbf{h}_l \in \mathbb{R}^{n_l}$ and the output of the ReLU DNN are specified in (4a) and (4b), respectively.

$$\mathbf{h}_l = \sigma[\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l] \quad \forall l \in \{1, \dots, L\} \quad (4a)$$

$$f(\mathbf{z}) = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \quad (4b)$$

Based on the definitions above, we propose Theorem 3.1.

Theorem 3.1. Let us define functions $c : \mathbb{R}^{n_0} \rightarrow \mathbb{R}$ and $g : \mathbb{R}^{n_0} \rightarrow \mathbb{R}^e$, where $e \in \mathbb{N}$. Let us also consider the ReLU DNN f as defined in Definition 2 with a single output i.e.,

$n_{L+1} = 1$. Thus, we can define a cost minimisation optimisation problem of the form below:

$$\min_{\mathbf{z} \in \mathbb{R}^{n_0}} c(\mathbf{z}) + f(\mathbf{z}) \quad (5a)$$

$$s.t. \quad g(\mathbf{z}) \leq 0 \quad (5b)$$

For some $k \in \mathbb{N}$ and $k \leq L$, if $\mathbf{W}_l \geq 0$ (element-wise) $\forall l \in \{k+1, \dots, L+1\}$, then the optimisation problem can be reformulated as

$$\min_{\mathbf{z} \in \mathbb{R}^{n_0}} c(\mathbf{z}) + f(\mathbf{z}) \quad (6a)$$

$$s.t. \quad g(\mathbf{z}) \leq 0 \quad (6b)$$

$$f(\mathbf{z}) = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \quad (6c)$$

$$\mathbf{h}_0 = \mathbf{z} \quad (6d)$$

$$\mathbf{h}_l = \sigma[\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l] \quad \forall l \in \{1, \dots, k-1\} \quad (6e)$$

$$\mathbf{h}_l \geq \mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l \quad \forall l \in \{k, \dots, L\} \quad (6f)$$

$$\mathbf{h}_l \geq 0 \quad \forall l \in \{k, \dots, L\} \quad (6g)$$

Proof. Let us define a simplified version of the problem, that is, a ReLU DNN with a single neuron:

$$\min_{z \in \mathbb{R}} f(z) \quad (7a)$$

$$s.t. \quad g(z) \leq 0 \quad (7b)$$

$$f(z) = \max(az, 0) \quad (7c)$$

The standard LP reformulation of this problem is

$$\min_{z \in \mathbb{R}} f(z) \quad (8a)$$

$$s.t. \quad g(z) \leq 0 \quad (8b)$$

$$f(z) \geq az, \quad f(z) \geq 0 \quad (8c)$$

Because the problem aims to minimise $f(z)$, this reformulation will ensure $f^*(z) = az$ or $f^*(z) = 0$. If it was a maximisation problem, $f(z)$ would be unbounded. Thus, replacing max (or ReLU) constraints with linear inequalities requires the output of the neuron to be minimised.

Now let us consider a problem with nested maximum constraints, such as a ReLU DNN:

$$\min_{\mathbf{z}, \mathbf{u} \in \mathbb{R}^2} f(\mathbf{z}) \quad (9a)$$

$$s.t. \quad g(\mathbf{z}, \mathbf{u}) \leq 0 \quad (9b)$$

$$f(\mathbf{z}) = \max\left(\sum_{j=1}^2 a_j z_j, 0\right) \quad (9c)$$

$$z_i = \max\left(\sum_{j=1}^2 b_{ij} u_j, 0\right) \quad \forall i \in \{1, 2\} \quad (9d)$$

Here, $a \in \mathbb{R}^{2 \times 1}$ and $b \in \mathbb{R}^{2 \times 2}$ are the weight matrices of the ReLU DNN layers. A reformulation similar to the one demonstrated above would look as follows:

$$\min_{\mathbf{z}, \mathbf{u} \in \mathbb{R}^2} f(\mathbf{z}) \quad (10a)$$

$$s.t. \quad g(\mathbf{z}, \mathbf{u}) \leq 0 \quad (10b)$$

$$f(\mathbf{z}) \geq \sum_{j=1}^2 a_j z_j, \quad f(\mathbf{z}) \geq 0 \quad (10c)$$

$$z_i \geq \sum_{j=1}^2 b_{ij} u_j, \quad z_i \geq 0 \quad \forall i \in \{1, 2\} \quad (10d)$$

If $a_1, a_2 \geq 0$, the optimisation problem aims to minimise z_1 and z_2 binding the max constraint. However, if $a_1 < 0$ or $a_2 < 0$, the problem aims to maximise z_1 or z_2 , respectively. Thus, the associated max constraint becomes unbounded. Solving this problem will provide solutions different from the original problem. Applying this argument, we justify the reformulation presented in (6). In essence, we are “convexifying” a section of the hidden layers (from k to L) of a ReLU DNN by constraining their weight matrix. Owing to this convexification, the convexified section can be replaced with LP reformulations within minimisation problems, whereas the non-convexified section (from 1 to $k - 1$) cannot be linearised. We note here that the ReLU DNN can only be convexified in reverse, i.e., from output layer onwards by continuously constraining the weights to be non-negative. Therefore, a ReLU DNN can be completely replaced by an LP reformulation within optimisation problems, with guaranteed tightness, if it satisfies two conditions: 1) the output of the ReLU DNN is minimised and 2) the weight matrix of all layers barring the first ReLU layer must be non-negative matrices. $\square$

Definition 3. A ReLU DNN f , as defined in Definition 2 with a single output (i.e., $n_{L+1} = 1$), is called a *cvxd ReLU DNN* if all weight matrices from the second layer onward are element-wise non-negative, i.e., $\mathbf{W}_l \geq 0$ (element-wise) for all $l \in \{2, \dots, L + 1\}$.

Clearly, a cvxd ReLU DNN can be equivalently represented through an LP reformulation, leveraging the results established in Theorem 3.1. Although weight-constrained ReLU DNNs have been extensively studied in the literature, particularly for their monotonicity properties, we are the first to propose an LP reformulation for them by leveraging their convexity properties.

ReLU DNNs are universal approximators, but in applications such as finance and health-care, preserving monotonicity is crucial. One way to achieve this is to constrain the sign of the weights in ReLU DNNs. However, these constraints affect the expressiveness of ReLU DNNs, limiting them to learning and accurately representing only convex and monotonic functions, as demonstrated in [23]. Thus, cvxd ReLU DNNs can accurately learn convex and monotonically increasing functions. This limitation poses a caveat when applying this formulation, as it trades accuracy and expressiveness for tractability and tightness, as demonstrated by our results in Section 4. Despite this limitation, the results in Section 4 demonstrate that cvxd ReLU DNNs can effectively capture nonlinear and nonconvex monotonically increasing flexibility cost functions for aggregator’s operation. Consequently, cvxd ReLU DNNs may also be suitable for learning nonlinear fuel cost functions or CO₂ emissions, which inherently increase monotonically.

3.2. SOTA ReLU DNN reformulations

The authors in [19] proposed two LP reformulations for ReLU activation functions: PCAR and PCTAR. Both methods are based on the same core idea as our proposed ReLU DNN reformulation, namely that a ReLU neuron can be linearised if its output is minimised within the optimisation problem.



Figure 2: ReLU function feasible space for LP reformulations adapted from [19]

In our approach, we leverage this principle by first verifying whether the optimisation objective involves minimising the output of the ReLU DNN. If so, we propagate this minimisation behaviour throughout the entire network by constraining all weight matrices of the ReLU DNN to be non-negative. This structural enforcement ensures that the outputs of all intermediate neurons are also minimised.

In contrast, PCAR and PCTAR do not impose any structural constraints on the DNN weights $\mathbf{W}_l$ for $l \in 1, \dots, L$. Instead, they introduce penalty functions directly within the optimisation model to implicitly encourage the minimisation of neuron outputs. This makes them broadly applicable to generic ReLU DNNs with arbitrary weights. PCTAR is a further relaxation of PCAR, where the feasible space of each ReLU neuron is restricted to a triangular region, as illustrated in Fig. 2, in order to improve tractability.

In the following, we briefly describe the mathematical formulations of PCAR and PCTAR under the assumption that the output of the ReLU DNN is minimised in the optimisation problem. For more general formulations, we refer the reader to the original work [19]. We retain the notation introduced in Section 3.1 with $n_{L+1} = 1$, and define $\boldsymbol{\alpha}_l \in \mathbb{R}^{n_l}$ as the vector of penalty coefficients for layer l , and $LB, UB \in \mathbb{R}$ as the lower and upper bounds of each ReLU neuron, respectively. The formulations for PCAR and PCTAR are specified in Eqs. (11a)–(11f) and Eqs. (12a)–(12g).

$$\min_{\mathbf{z} \in \mathbb{R}^{n_0}} c(\mathbf{z}) + f(\mathbf{z}) + \sum_{l=1}^L \boldsymbol{\alpha}_l^T \mathbf{h}_l \quad (11a)$$

$$\text{s.t. } g(\mathbf{z}) \leq 0 \quad (11b)$$

$$f(\mathbf{z}) = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \quad (11c)$$

$$\mathbf{h}_0 = \mathbf{z} \quad (11d)$$

$$\mathbf{h}_l \geq \mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l \quad \forall l \in \{1, \dots, L\} \quad (11e)$$

$$\mathbf{h}_l \geq 0 \quad \forall l \in \{1, \dots, L\} \quad (11f)$$

$$\min_{\mathbf{z} \in \mathbb{R}^{n_0}} c(\mathbf{z}) + f(\mathbf{z}) + \sum_{l=1}^L \boldsymbol{\alpha}_l^T \mathbf{h}_l \quad (12a)$$

$$\text{s.t. } g(\mathbf{z}) \leq 0 \quad (12b)$$

$$f(\mathbf{z}) = \mathbf{W}_{L+1} \mathbf{h}_L + \mathbf{b}_{L+1} \quad (12c)$$

$$\mathbf{h}_0 = \mathbf{z} \quad (12d)$$

$$\mathbf{h}_l \geq \mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l \quad \forall l \in \{1, \dots, L\} \quad (12e)$$

$$\mathbf{h}_l \leq k_1 [\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l] + k_2 \quad \forall l \in \{1, \dots, L\} \quad (12f)$$

$$\mathbf{h}_l \geq 0 \quad \forall l \in \{1, \dots, L\} \quad (12g)$$

From the objective functions of both reformulations, Eqs. (11a) and (12a), additional penalty terms can be observed compared with our proposed reformulation. In the case of PCTAR, an additional constraint (12f) is introduced to further restrict the feasible space of each ReLU neuron to a triangular region, as illustrated in Fig. 2a. This constraint is defined by parameters $k_1 = \frac{UB}{UB-LB}$ and $k_2 = \frac{UB-LB}{UB-LB}$, and aims to improve the tractability by relaxing the feasible space.

3.3. ReLU DNN training strategy

In this section, we describe the training strategy for ReLU DNNs to learn the nonlinear and nonconvex relationship presented in Eq. (1c). We note here that despite its nonconvex nature, our aim is to demonstrate that cvxd ReLU DNNs may sufficiently capture the relationship to obtain reasonable quality solutions. The defined function has four inputs and one output, and is invalid if $x_t^i \geq \tilde{x}_t^i$. It should be noted that two sets of ReLU DNNs are trained using the strategy described below: 1) **cvxd ReLU DNN**, which is implemented using the proposed formulation as specified in Section 3.1; 2) **Unconstrained (UC) ReLU DNN**, which is implemented using the other formulations.

Let $\mathcal{D} \in \mathbb{R}_+^{N \times 4}$ denote the input dataset with $N_D \in \mathbb{N}$ data points. These points are randomly generated such that an input point $\mathbf{z} = \{x_t^i, \tilde{x}_t^i, q_t^i, r_t^i\}$ satisfies $\underline{\mathbf{z}} \leq \mathbf{z} \leq \bar{\mathbf{z}}$ and $z_1 < z_2$. Here, $\underline{\mathbf{z}}$ and $\bar{\mathbf{z}}$ are the lower and upper bounds of $\mathbf{z}$. These are then substituted in Eq. (1c) to obtain the corresponding output dataset $\mathcal{O} \in \mathbb{R}_+^N$, so the combined dataset is denoted as $\{\mathcal{D}, \mathcal{O}\}$.

To expedite the training process, we applied min-max normalisation to the combined dataset, normalising the input and output values to a range of $[0, 1]$. The resulting normalised combined input and output datasets, denoted as $\{\mathcal{D}, \mathcal{O}\}'$, were then randomly divided into training and validation sets in an 80:20 ratio. We trained the ReLU DNNs using the Adam optimiser with a learning rate of 0.0001 while applying the mean squared error (MSE) loss metric. The models were trained for 1000 epochs with a batch size of 1000. The validation set was used to monitor the training process and prevent overfitting.

4. Simulation study and results

In this section, we describe the simulation studies implemented to evaluate the performance of the proposed reformulation in Section 3.1 compared to SOTA methods.

4.1. Simulation study assumptions

In this study, we simulate scenarios in which Energinet only procures mFRR up-regulation capacity. This is justified by the fact that, historically, Energinet has rarely procured mFRR down-regulation capacity. This is evidenced by the average hourly volumes procured between August 2023 and April 2025, where approximately 390 MW of up-regulation capacity was acquired, compared to only 35 MW of down-regulation capacity [24]. In the future, these trends may change by evolving policy developments, such as the Nordic Balancing Model [25], a collaborative initiative among Nordic TSOs aimed at combining ancillary services market. Nonetheless, under current market conditions, aggregators may find limited economic or operational value in participating in down-regulation mFRR capacity markets.

Moreover, including down-regulation in the optimisation model significantly increases computational burden, particularly for Gurobi ML and PWL surrogate formulations, as it effectively doubles the number of binary variables by adding surrogates for both $\lambda_t^{P,u}$

and $\lambda_t^{P,d}$. We would like to note here that for the PCAR, PCTAR and the proposed formulations this is not as burdensome because of their LP nature. Thus, to maintain tractability for PWL and Gurobi ML models, we exclude down-regulation from our simulation study by setting $x_t^d = 0, \lambda_t^{P,d} = 0, \forall t \in \mathcal{T}$. As a result, the total purchase cost equals the purchase cost of up-regulation flexibility $\lambda_t^P = \lambda_t^{P,u}, \forall t \in \mathcal{T}$.

Furthermore, we assume that the mFRR up-regulation service is always fully activated, i.e., $a_t^\uparrow = 1, \forall t \in \mathcal{T}$, and the imbalance price is set to zero, i.e., $\lambda_t^B = 0, \forall t \in \mathcal{T}$. These assumptions effectively eliminate the influence of the activation market from our analysis, thereby isolating the value derived by the aggregator from participation in the mFRR capacity market alone. This simplification does not affect the core of our analysis, as the activation-related parameters primarily impact the net up-regulation price, given by $\lambda_t^U + a_t^\uparrow \lambda_t^B, \forall t \in \mathcal{T}$. Since Section 4.3 already examines the effect of varying price levels on model performance, the exclusion of imbalance pricing does not limit the validity or generality of the study's conclusions.

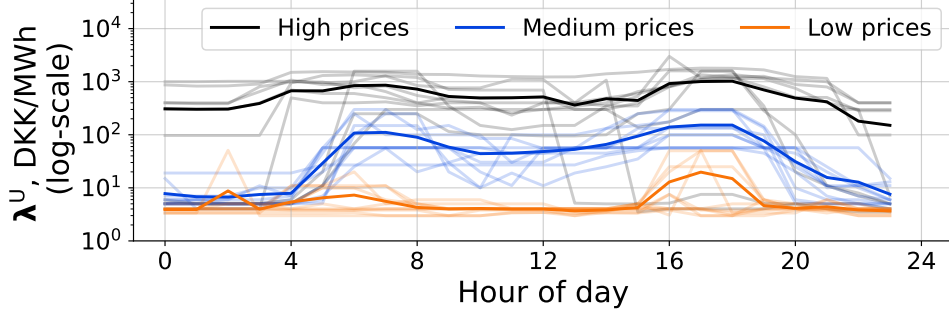
Regarding the rebound effect, our modelling approach assumes that any flexibility deployed in an interval results in a full and deterministic loss of future flexibility. This reflects a conservative, worst-case scenario that incentivises the optimiser to be careful when selecting intervals for participation. In particular, it eliminates the possibility of “free” scheduling in future intervals with low activation, thereby enforcing a realistic trade-off between present and future flexibility deployment. This assumption adds behavioural realism to the optimisation framework without increasing its computational complexity.

4.2. Optimisation parameters

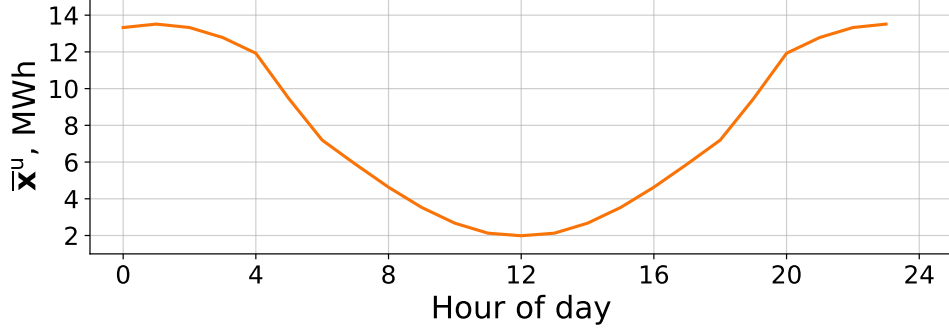
In this section, we describe the market and prosumer cluster parameters used within the optimisation model. As described in Section 2.1, we assume that the aggregator participates in the European mFRR capacity market, specifically in the DK1 region of Denmark, which is operated by Energinet. This market is operated one day in advance and reserves tertiary FCAS for every hour the next day. Thus, the optimisation horizon is $T = 24$ hours.

For this study, we obtained real market price scenarios for mFRR capacity market using the API of the Energinet data service [24]. We separated the price scenarios into low, medium, and high prices and selected ten scenarios in each of the above categories. This was done to study the influence of market prices on the tractability and performance of the approximation methods of $\lambda^{P,i}$ or in this case study $\lambda^{P,u}$. The price profiles for these categories are illustrated in Fig. 3(a) with the mean price profile highlighted. We would like to note that 90% of the prices are below 10 DKK/MWh, making low-price scenarios highly likely and consequently the critical case. However, because of the nature of this market, when prices are high, they are orders of magnitude higher, as demonstrated by the log-y scale in Fig. 3(a). Therefore, studying the performance of the reformulations for these three price categories is crucial for this application.

The data of the prosumer cluster was generated randomly based on our previous work [20]. The maximum up-flexibility obtainable profile ($\bar{x}^u$) used in this study is shown in Fig. 3(b). Note that this profile follows a sinusoidal pattern to account for diurnal factors, as demonstrated by the work done in [26]. The cross-temporal matrices associated with activated up-regulation and updated maximum up-regulation flexibility ($A^{u,\uparrow}$) is a randomly generated lower triangular matrix with diagonal elements equal to zero. The price elasticity of prosumers can be used as a proxy for $A^{u,\uparrow}$, which can be obtained using real-



(a) mFRR capacity market price by category



(b) Maximum up-regulation flexibility obtainable from the prosumer cluster

Figure 3: Optimisation problem parameters

world data, as shown by [27]. Furthermore, the total flexibility lost in future intervals must be less than or equal to the flexibility reserved in the current interval t , that is, $\sum_{i \in \mathcal{T}} A_{i,t}^{u,\uparrow} \leq 1, \forall t \in \mathcal{T}$. We ignore the rest of the cross-elasticities as down-regulation market is removed from the study.

For the prosumer responsiveness model, the shaping parameters q_t^u, r_t^u were randomly generated and fixed for each hour $t \in \mathcal{T}$. These parameters can be derived by fitting a saturation curve to real-world data using regression [28]. For the two-dimensional PWL model [22], we set the number of pieces to $N_P = 4$ for both inputs.

For the PCTAR method, bounds for the neurons were set as $LB = -10, UB = 10$ to account for potential outliers, given that training with min-max normalisation typically keeps neuron inputs and outputs within ± 1 . To train the ReLU DNNs, $N_D = 3 \times 10^5$ data points were generated following the methodology in Section 3.3. We note that the average CPU training time for all the ReLU DNN architectures used in this case study was 587 seconds.

4.3. Tractability, accuracy and solution quality analysis

In this section, we evaluate the tractability, accuracy, and solution quality achieved using different $\lambda^{P,u}$ surrogate models, based on the mathematical models presented in Section 2.2. For this, we trained UC and cvxd ReLU DNNs with three hidden layers consisting of 10, 20, and 10 neurons, respectively, as discussed in Section 3.3. These trained neural networks are used to represent $\lambda_t^{P,u}, \forall t \in \mathcal{T}$ within our optimisation model using the Gurobi ML package [16], PCTAR, PCAR, and the proposed reformulations. In addition, we employed the PWL method as a benchmark for evaluating these approaches based on neural networks. The formulated optimisation models were then solved using Gurobi[®] 11.0 [29] on a PC with 16GB RAM and an Intel[®] i7 16-core processor. We set the stopping criteria to be 3600 seconds or MIP Gap of 1%, whichever came first.

Since PCAR and PCTAR are penalty-driven approaches, we tested various penalty values during optimisation, as detailed in Section 4.5. For comparison, we selected the penalty value that maximised the realised profit and minimised the runtime for each price scenario.

4.3.1. Computational performance

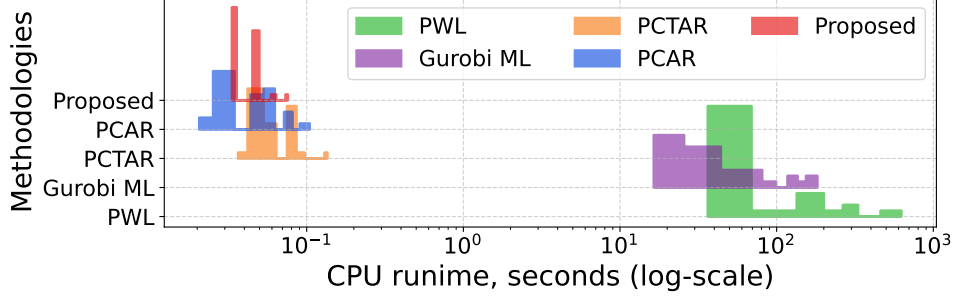


Figure 4: CPU runtime histogram for all surrogate models across all simulated scenarios

Figure 4 shows a histogram of the CPU runtime for all price scenarios solved using various surrogate models. CPU runtimes provide insight into the amount of computer resources being used, which in turn affects the operational costs for the aggregator. Please note that the x-axis of this plot is on a log scale. As expected, we can see that the LP reformulations PCTAR, PCAR, and the proposed reformulation significantly outperform the PWL and Gurobi ML reformulations in terms of speed and CPU resource usage, by a factor ranging from hundreds to thousands. This is because parallel processing is used by Gurobi[®] for branch-and-bound algorithms to solve MIPs, which requires more computational effort.

Table 1: Tractability analysis under different price scenarios

Price case	Surrogate model	Mean realised profit DKK	Mean run-time	Mean MIP gap %
High	PWL	50402	1 s	1
	Gurobi ML	50253	3 s	1
	PCTAR	50861	44 ms	0
	PCAR	50861	26 ms	0
	Proposed	50670 (0.4%)	34 ms	0
Medium	PWL	3910	6 s	1
	Gurobi ML	3678	8 s	1
	PCTAR	3870	44 ms	0
	PCAR	3870	31 ms	0
	Proposed	3894 (0.4%)	31 ms	0
Low	PWL	229	29 s	0
	Gurobi ML	171	15 s	1
	PCTAR	-514	65 ms	0
	PCAR	-514	48 ms	0
	Proposed	226 (1.3%)	35 ms	0

From Table 1, we note that all models are relatively fast in the high price scenarios, followed by medium price scenarios. This can be explained in Fig. 3(a), where the high and medium price scenarios are orders of magnitude larger and more volatile than the low price cases. Thus, the decision for the optimiser is fairly straightforward, as it is easier for the branch-and-bound algorithm to reject sub-optimal solutions. Conversely, in the case of low-price scenarios, the purchase cost and mFRR prices are most likely comparable making it harder for the optimiser to identify and reject sub-optimal solutions.

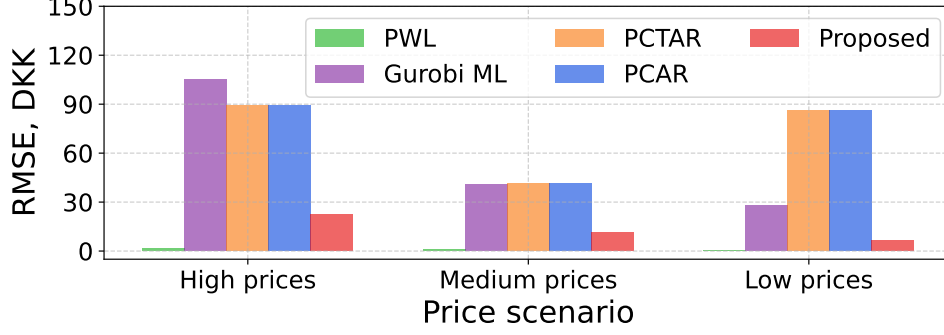


Figure 5: RMSE between actual and surrogate model $\lambda^{P,u}$ across different price scenarios

4.3.2. Solution quality and surrogate model accuracy analysis

In Table 1, the realised profit is calculated by substituting the optimal decision vectors $\mathbf{x}^u$ and $\tilde{\mathbf{x}}^u$ into Eqs.(1c) and (2a), reflecting the true profit based on the actual post-optimisation cost of up-regulation flexibility purchase. PCTAR and PCAR yield the highest average profit in high price scenarios, while PWL performs best under medium and low prices. As shown in Fig. 5, PWL also delivers the most accurate cost estimates, which is expected given its ability to closely approximate nonlinear functions with enough segments. This is further substantiated by Fig. 6(a), where the estimated cost of the up-regulation flexibility is very close to the actual cost. However, as discussed in Section 4.3.1, the PWL method is significantly more computationally expensive than our proposed method, and there also exist practical limitations, as discussed in Section 4.6, which may make it unsuitable in the real world.

Gurobi ML employs an MIP-based reformulation of the ReLU DNN, which is expected to provide a tight embedding of the network within the optimisation model. Notably, it uses the UC ReLU DNN to learn the $\lambda^{P,u}$ relationship, which achieved a lower training RMSE compared to the cvxd ReLU DNN used in our proposed formulation, as demonstrated using Figs. 7(a) and (b). Therefore, it was reasonable to expect Gurobi ML to exhibit superior realised RMSE. However, it deviated from this expected trend, underperforming our model in terms of realised accuracy. This deviation was also observed across other network architectures, as discussed in Section 4.4. The exact reason remains unclear and may be attributed to the proprietary nature of Gurobi[®] solution algorithms.

PCAR and PCTAR often lead to losses in low price scenarios, both the most probable and critical, due to substantial mis-estimation of the purchase cost of the up-regulation flexibility. This is reflected in the high RMSE values in Fig. 5. While these approaches perform adequately in high- and medium-price settings, they struggle when market prices are low. Figures 6(d) and (e) illustrates this issue for a representative low-price scenario. Despite bidding maximum flexibility at most intervals, PCAR and PCTAR estimate the purchase cost as 0 DKK, which leads to actual costs exceeding revenues and resulting in losses. This behaviour stems from the inability of penalty functions to consistently enforce the minimisation of all the neurons output. In many cases, the optimiser can decouple intermediate neurons from the ReLU DNN feed-forward equations, setting them freely to force low estimated costs while bidding maximum flexibility. As a result, the trained ReLU DNN's behaviour cannot always be reliably replicated, an issue further discussed in Section 4.4.

Our proposed model consistently achieves the second highest realised profit across all scenarios, remaining within approximately 1% of the top-performing method, as indi-

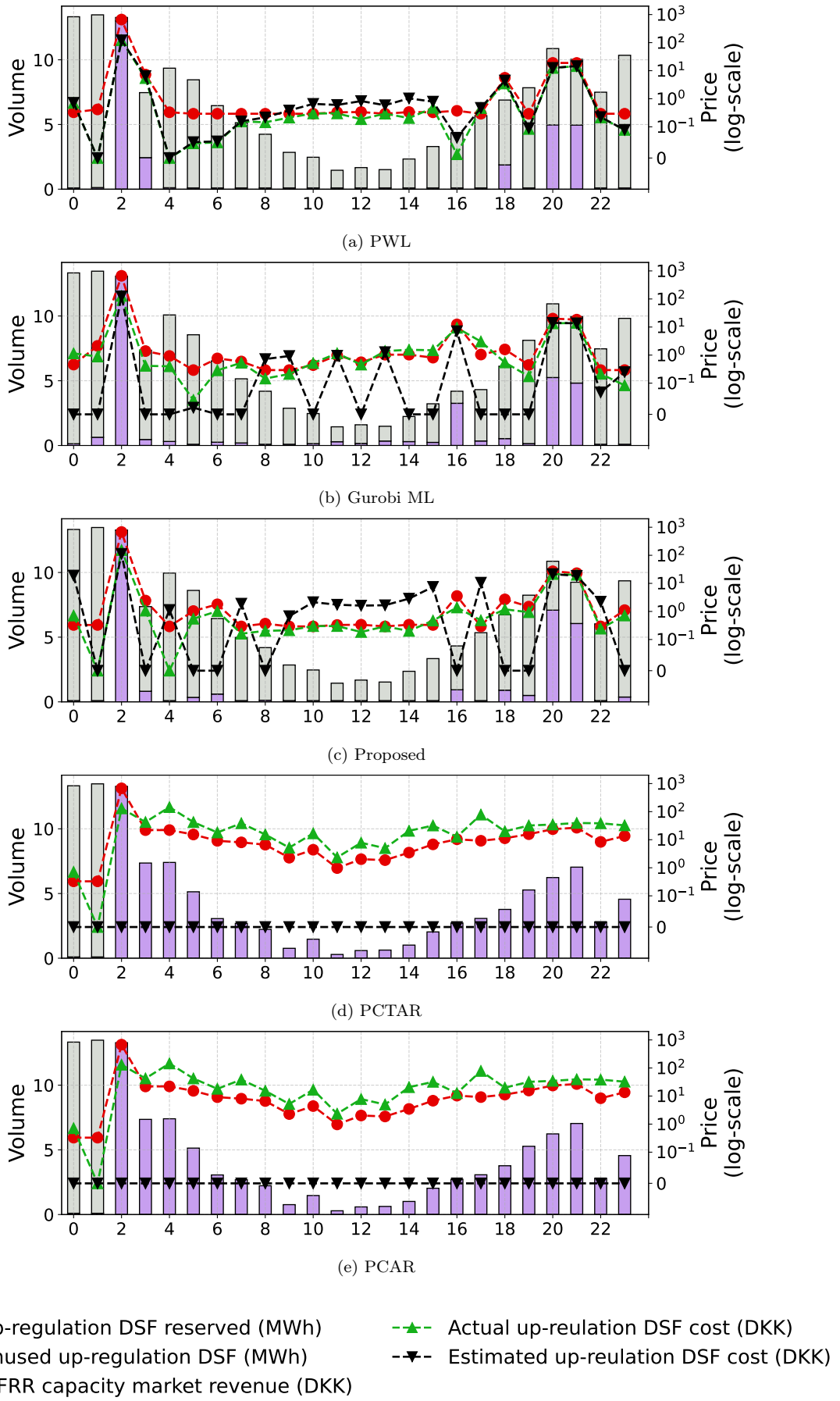


Figure 6: Optimisation results for one low price instance

cated by the percentage differences in parentheses in Table 1. It also ranks second in realised RMSE as shown in Fig. 5, outperforming Gurobi ML as noted above. While our formulation employs cvxd ReLU DNNs to learn $\lambda^{\mathbf{P},\mathbf{u}}$, architectures typically more suited to capturing convex nonlinearities, they still perform well in this nonconvex setting, delivering solid accuracy and solution quality at a fraction of computational speed.

It is interesting to note from Figs. 6(b) and (c) that both Gurobi ML and the proposed method are able to reasonably estimate the up-regulation flexibility cost when the reserved up-regulation flexibility volume is relatively high. However, for lower volumes, both methods tend to underestimate or overestimate the cost. This discrepancy may stem from the underlying ReLU DNNs struggling to accurately learn the cost relationship at lower volume levels. Consequently, the training performance of the embedded ReLU DNNs plays a significant role in the ultimate bidding behaviour and effectiveness of these approaches. This highlights the broader challenge of integrating learned models within optimisation frameworks, especially under data regimes where performance generalisation is critical.

4.4. ReLU DNN length and width sensitivity analysis

In this section, we evaluate the tractability, accuracy and tightness of the ReLU DNN-based approximation methods across varying network architectures. Two sets of cases are considered: one in which the number of neurons (i.e., the width) is varied while keeping the number of hidden layers (length) fixed, and another where the length is varied with a fixed total width. We focus exclusively on the ten low price scenarios, as they represent the most likely and critical operating conditions.

As outlined in Section 4.3, for each architecture, we train two sets of ReLU DNNs: the UC ReLU DNN embedded using the Gurobi ML, PCTAR, and PCAR reformulations, and the cvxd ReLU DNN embedded using our proposed reformulation. Additionally, we perform a penalty sensitivity analysis for both PCAR and PCTAR across all architectures, as detailed in Section 4.5. Recall that the baseline ReLU DNN architecture used in Section 4.3 consists of three hidden layers with 10, 20, and 10 neurons, respectively, resulting in a total width of 40.

For the width sensitivity analysis, we fix the length at three layers and vary the total width to 20, 80, 200, and 400 neurons. The neurons are distributed across the layers in the same proportion as in the baseline model.

For the length sensitivity analysis, we fix the total width to 40 neurons (total width of the baseline ReLU DNN) and vary the number of layers using the following architectures: 1) single layer with 40 neurons; 2) two layers with 20 neurons each; 3) four layers with 5, 15, 15, 5 neurons; and 4) five layers with 2, 8, 20, 8, 2 neurons.

4.4.1. Tractability and solution quality analysis

Tables 2 and 3 present the results of the width and length sensitivity analyses, respectively. As expected, across all ReLU DNN architectures, the LP-based surrogate models (PCAR, PCTAR, and the proposed method) consistently outperform Gurobi ML’s MIP-based formulation in terms of computational efficiency. Notably, the width of the ReLU DNN has a substantially greater impact on computational time than its length. This is because increasing the number of neurons directly increases the number of constraints in all surrogate models, thus increasing computational complexity. For Gurobi ML in particular, the addition of neurons leads to more integer variables, which significantly increases runtime due to the exponential growth in the complexity of solving MIPs.

Table 2: Width sensitivity analysis

Width	Surrogate model	Mean realised profit DKK	Mean run-time	Mean MIP gap %
20	Gurobi ML	227	2 s	0
	PCAR	-214	15 ms	0
	PCTAR	-214	8 ms	0
	Proposed	231	16 ms	0
80	Gurobi ML	221	3600 s	127
	PCAR	-36	78 ms	0
	PCTAR	-36	181 ms	0
	Proposed	237	56 ms	0
200	Gurobi ML	115	3600 s	884
	PCAR	-147	549 ms	0
	PCTAR	-147	921 ms	0
	Proposed	239	259 ms	0
400	Gurobi ML	-18	3601 s	236*
	PCAR	-9	3 s	0
	PCTAR	-22	6 s	0
	Proposed	238	748 ms	0

Table 3: Length sensitivity analysis

Length	Surrogate model	Mean realised profit DKK	Mean run-time	Mean MIP gap %
1	Gurobi ML	234	4 s	1
	PCAR	-359	13 ms	0
	PCTAR	-359	27 ms	0
	Proposed	239	18 ms	0
2	Gurobi ML	233	87 s	1
	PCAR	-118	38 ms	0
	PCTAR	-118	70 ms	0
	Proposed	237	30 ms	0
4	Gurobi ML	209	53 s	1
	PCAR	106	37 ms	0
	PCTAR	106	59 ms	0
	Proposed	234	27 ms	0
5	Gurobi ML	156	3600 s	55
	PCAR	47	12 ms	0
	PCTAR	47	32 ms	0
	Proposed	219	35 ms	0

In contrast, the length of the ReLU DNN appears to have minimal impact on the computational performance of the LP-based surrogate models. However, for Gurobi ML, the mean runtime generally increases with network length, except in the case of length 4, where a reduction is observed. For length 5, the mean runtime abruptly spikes to 3600 seconds, suggesting potential scalability issues with deeper networks. Since the exact workings of the Gurobi ML’s algorithm are not publicly disclosed, it is difficult to pinpoint the exact cause of this irregularity.

In both Tables 2 and 3, the mean realised profit is re-evaluated post-optimisation, as described in Section 4.3.2. The results clearly demonstrate that our proposed formulation consistently yields the highest profit across all ReLU DNN architectures. In contrast, the PCAR and PCTAR models generally incur losses due to the mis-estimation of the up-regulation flexibility cost, as discussed in Section 4.3.2. The only instances where these models make a profit occur when the number of hidden layers exceeds three, suggesting that the length of the ReLU DNN has some influence on the quality of their solution. However, even in these cases, the realised profit is less than half of what is achieved by our proposed method.

Meanwhile, Gurobi ML shows relatively better performance than PCAR and PCTAR in terms of realised profit, but its performance deteriorates significantly with increasing network width and length. This is primarily due to its poor scalability, as the solver struggles to find high-quality feasible solutions within the 3600-second time limit. This is evident from the increasing mean MIP gap percentages. Notably, for the architecture with a width of 400 neurons, Gurobi ML was unable to solve 8 out of 10 price scenarios within the time limit, and the reported mean MIP gap is based only on the two solvable cases, as indicated by *. These results indicate that Gurobi ML does not tractably scale to wider ReLU DNNs, which poses challenges for applications requiring larger neural network architectures.

Overall, these findings demonstrate that our proposed ReLU DNN reformulation significantly outperforms existing SOTA methods in both computational efficiency and solution quality, across all tested ReLU DNN architectures.

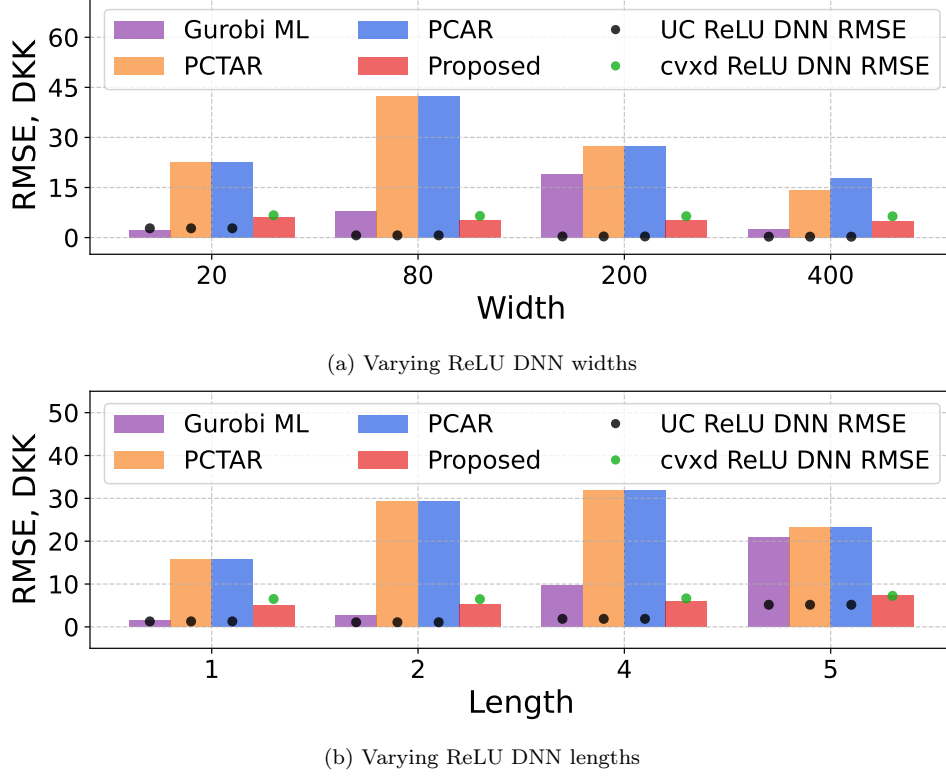


Figure 7: RMSE between actual and surrogate model $\lambda^{P,u}$ across varying ReLU DNN architectures, along with their corresponding training RMSE

4.4.2. Surrogate model accuracy and tightness analysis

Figures 7(a) and (b) present the RMSE between the true up-regulation flexibility price, computed post-optimisation, and the surrogate price generated within the optimisation model, across various ReLU DNN architectures. In addition, the training RMSE is reported for each architecture. It is important to note that for each ReLU DNN architecture, we simulated the ten low price scenario over a 24-hour period, resulting in 240 data points per architecture, which were used to compute these RMSE values.

The results demonstrate that our proposed ReLU DNN reformulation consistently achieves a lower RMSE across all tested architectures. In contrast, both PCAR and PCTAR exhibit the highest RMSE, indicating the poorest accuracy among the compared methods. Its accuracy also varies inconsistently across the ReLU DNN architectures, making it hard to comment on its behaviour with increasing length and width. The Gurobi ML model performs well for smaller and shallower ReLU DNNs (e.g., width of 20 and depth below 3), but its accuracy deteriorates as network complexity increases. An exception is observed at a width of 400, where Gurobi ML achieves low RMSE; however, as noted in Section 4.4.1, only two out of ten cases were feasibly solved in this configuration, which may skew the RMSE results. These variations in the accuracy of the surrogate model directly impact the mean realised profit, as discussed in Section 4.4.1, where higher accuracy corresponds to greater profit in most cases.

Further supporting the robustness of our approach, we observe that the proposed model consistently achieves RMSE values close to the training RMSE of the cvxd ReLU DNN. This behaviour clearly demonstrates the tightness of our formulation, as established in Theorem 3.1. For shallower and smaller DNNs, the Gurobi ML model achieves RMSE values close to the training RMSE of the UC ReLU DNN. However, as the network be-

comes deeper and more complex, this tightness is lost, and the resulting RMSEs become significantly higher than the corresponding UC ReLU DNN training RMSE. This behaviour is unexpected, as Gurobi ML is presumed to be based on an MIP reformulation of ReLU DNNs, which should, in principle, maintain tightness [16].

As expected, both PCAR and PCTAR yield RMSE values that are substantially higher than those of the UC ReLU DNNs they approximate, clearly indicating that they are not tight ReLU DNN reformulations. The reasoning behind this behaviour is detailed in Section 4.3.2, and stems from the structural uncoupling of ReLU DNN neurons from feed-forward equations, leading to reduced fidelity. These observations reinforce the conclusion that, unlike other SOTA techniques, our proposed model is not highly sensitive to the underlying ReLU DNN architecture, and consistently maintains both accuracy and tightness across scenarios.

Finally, we examine the effect of varying ReLU DNN architectures on the training RMSEs themselves. It is evident that the cvxd ReLU DNN produces relatively consistent RMSEs across all architectures, which can be attributed to its limited expressivity due to the non-negativity constraints on the hidden layer weight matrices. In contrast, UC ReLU DNNs show better performance as the network size increases, benefiting from their greater expressiveness. They are also highly accurate, achieving significantly lower RMSEs compared to the cvxd ReLU DNNs, demonstrating their strong learning capabilities. This highlights a core limitation of our proposed formulation: learning-based optimisation methods are inherently bounded by the quality of the learning models they embed. Thus, if the embedded model (e.g., the cvxd ReLU DNN) is suboptimal due to its training constraints, the overall performance of the OCL framework may suffer despite tight reformulations.

4.5. Penalty sensitivity analysis for PCAR and PCTAR

Because PCTAR and PCAR are penalty-driven approaches, we performed a penalty sensitivity analysis to determine the best penalty value for each price scenario and the ReLU DNN architecture. We considered two classes of penalty values: 1) constant penalty values applied to all neurons 0.01, 1, 10, 1000 and 2) layer-based penalty values $5^l, 2^l, 2^{-l}, 5^{-l}, 10^{-l}$ for all the neurons in the l^{th} hidden ReLU layer.

Figure 8 shows sample penalty sensitivity results for the three sets of price scenarios. The x-axis specifies the realised profit and the y-axis specifies the runtime for the scenario. The best penalty value is one which maximises realised profit and minimises runtime; hence, values closer to the bottom right corner of the graph. From the three examples, we can see that the best penalty value may change across cases and the profit obtained varies significantly with the penalty value chosen.

Furthermore, we identified the optimal penalty value across all cases for the high price, medium price, and low price scenarios: $5^{-l}, 2^{-l}, 1000$ for the PCTAR reformulation and $1, 5^{-l}, 1000$ for the PCAR reformulation, based on the study conducted in Section 4.3. Regardless of the variations in length and width discussed in Section 4.4, a penalty value of 1000 consistently yielded the best results for all architectures. This suggests a strong correlation between the penalty value and the underlying optimisation parameters. This may work well in this application where most of the price scenarios fall into the low price case. However, in other applications where these parameters are more evenly distributed, identifying the best penalty value may prove more challenging.

Across all cases, the realised RMSE of PCAR and PCTAR are generally comparable, except when the width is set to 400, where PCAR achieves a lower RMSE. This discrepancy

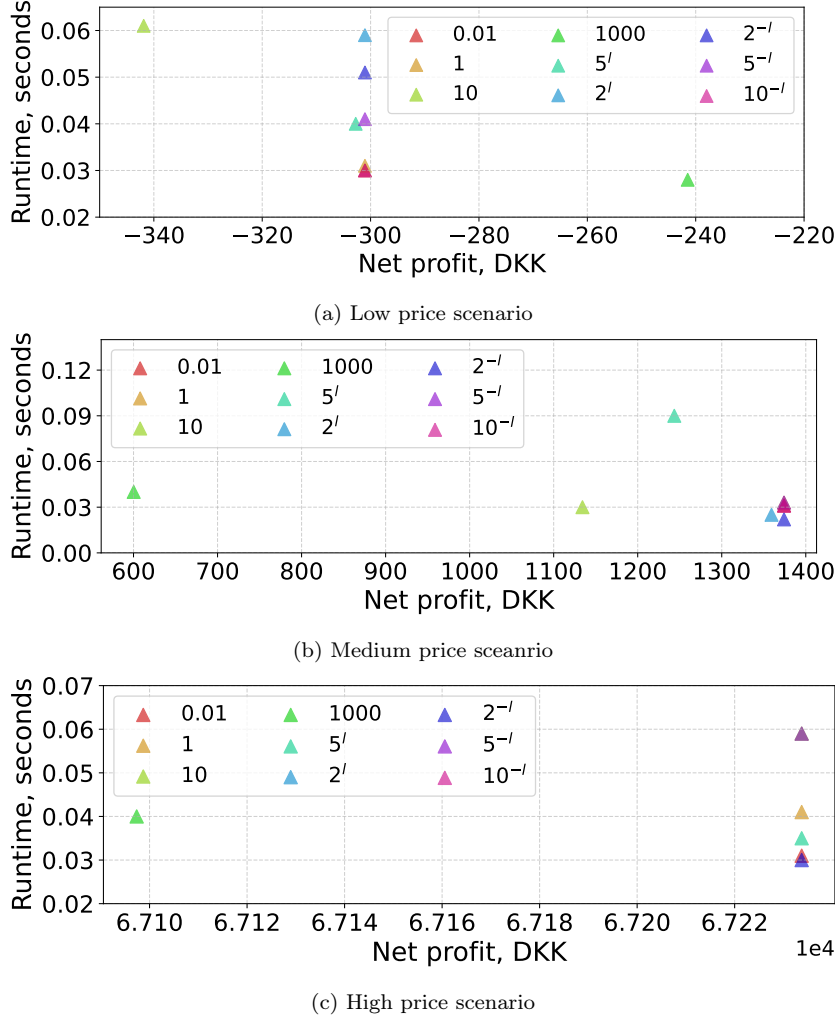


Figure 8: Sample penalty sensitivity analysis examples for each price scenario

arises from insufficient LB, UB selection in the PCTAR case, which did not fully capture the neuron outputs, leading to a worse realised profit value, as shown in Table 2. This highlights the additional complexity involved in tuning the parameters for the PCTAR method.

In summary, this section demonstrates that the performance of PCAR and PCTAR is highly sensitive to the choice of penalty values. Achieving competitive accuracy and runtime performance requires careful calibration, and improper tuning can lead to significant deterioration in both surrogate model fidelity and economic outcomes.

4.6. Discussion and practical considerations

In this section, we discuss the key insights obtained from this study and highlight practical considerations when implementing the various surrogate models.

The piecewise linear (PWL) approach is widely adopted in power system optimisation problems and demonstrated the best performance in our study, accurately representing prosumers' behaviour while delivering the most favourable economic outcomes (Section 4.3). Moreover, it solves the problem within 30 seconds, an acceptable runtime for day-ahead scheduling, raising the question of whether alternative surrogate models are necessary. As outlined in Sections 2.1 and 4.1, we have substantially simplified both the prosumer responsiveness and the aggregator's business model to maintain tractability for

the PWL and Gurobi ML formulations. In real-world settings, prosumer behaviour is influenced by numerous interdependent factors, making it infeasible to capture through closed-form expressions, thereby necessitating data-driven approaches. Additionally, aggregator participation in multiple markets (e.g., up- and down-regulation) introduces further modelling and computational challenges. These factors impose a significant computational burden, which ultimately makes the PWL approach impractical for real-time or large-scale aggregator operations.

Our results also highlight the potential of ReLU DNNs to effectively model nonlinear and nonconvex prosumer behaviour, even with relatively simple architectures (Section 4.4.2). This includes cvxd ReLU DNNs, which are theoretically well-suited for approximating convex functions. This suggests the need for further exploration, particularly as more data on prosumer behaviour becomes available. However, tractable reformulations are required to embed such models into optimisation problems. While the standard MIP reformulation, coupled with bound-tightening strategies (as in Gurobi ML), offers theoretical tightness guarantees and performs well for shallow networks, our findings in Section 4.4 reveal that deeper or more complex ReLU DNNs pose computational challenges and may even compromise tightness, resulting in worse economic outcomes. In such cases, LP-based reformulations, such as those proposed in [19] and in this work, may provide more viable alternatives under the right conditions.

Although the PCAR and PCTAR methods are presented as generic LP reformulations for ReLU DNNs applicable to a wide range of optimisation problems, they inherently trade off tightness and fidelity due to their penalty-driven structure. As a result, despite their generality, they tend to yield negative economic outcomes in our case study (Sections 4.3 and 4.4). A central challenge lies in tuning the penalty parameters, which are highly sensitive to both the scale of optimisation variables and the structure of the ReLU DNN (Sections 4.4 and 4.5). This sensitivity makes PCAR and PCTAR difficult to implement reliably in practice, as they offer no formal tightness guarantees for any fixed penalty setting—even if acceptable performance is observed in specific instances. Consequently, even if the original ReLU DNN captures complex system dynamics accurately, these surrogate reformulations may fail to reflect them within the optimisation problem.

By contrast, our proposed cvxd ReLU DNN effectively captures the nonlinear and nonconvex prosumer response function defined in this case study. When embedded in the aggregator’s profit maximisation problem using our LP reformulation, it achieves good quality solutions comparable to the PWL method at a fraction of the computational cost (Section 4.3). Moreover, it demonstrates consistent performance across a range of optimisation scales and ReLU DNN architectures. As such, cvxd ReLU DNNs, when paired with our proposed LP formulation, offer a promising pathway to improving computational efficiency while preserving solution quality within the OCL framework. That said, their applicability is limited by both the restricted expressiveness of cvxd ReLU DNNs and the ReLU DNN output minimisation structure required by our reformulation, confining them to specific problem classes. Nonetheless, they warrant further exploration as a viable surrogate modelling strategy to learn other nonlinear and nonconvex functions, such as CO₂ emissions or nonlinear generator cost functions, within modern power system optimisation problems.

Finally, our findings suggest that the learning behaviour of the ReLU DNN within the OCL framework plays a critical role in determining overall optimisation performance. As illustrated in Fig. 6(b)–(e) and discussed in Sections 4.3.2 and 4.4.2, the quality of the learned approximation significantly influences the surrogate model’s effectiveness when

embedded in the aggregator’s bidding strategy. While this observation may appear intuitive, it underscores a practical challenge: ensuring that the ReLU DNN is trained with sufficient fidelity and generalisation capability to accurately reflect underlying system dynamics. In real-world implementations, this requires careful consideration of training data quality, network architecture, and regularisation techniques, all of which are essential for realising the full benefits of the OCL framework.

5. Conclusions and future scope

This paper developed and evaluated an OCL-based framework for flexibility aggregators operating in the European ancillary service markets, with a focus on embedding ReLU DNNs into tractable optimisation models. We proposed a novel LP reformulation of ReLU DNNs whose weight matrices are constrained to be non-negative beyond the first layer (cvxd ReLU DNNs), which guarantees tightness when the ReLU DNN output is minimised.

Through extensive simulations involving real market data from Denmark’s mFRR market, we demonstrated that our proposed formulation achieves solution quality comparable to the widely used PWL approach, while significantly reducing computational time. Our approach remains efficient across a variety of ReLU DNN architectures and optimisation parameter scales. The proposed reformulation also maintains fidelity to the underlying cvxd ReLU DNN, avoiding the fidelity loss observed in penalty-driven approaches like PCAR and PCTAR, which require architecture-dependent and optimisation parameter-dependent penalty tuning. Our results also reveal that even the MIP-based Gurobi ML reformulation struggles to maintain tightness and computational scalability for deeper or wider ReLU DNNs, despite being theoretically exact.

While the proposed method is limited to applications where the learned ReLU DNN output is part of a minimisation objective and to functions which the cvxd ReLU DNN can effectively learn, it presents a promising methodology for surrogate modelling of complex system behaviour. Future work may explore extensions to broader market contexts (combination of both up- and down-regulation mFRR markets and combination with other markets such as aFRR), integration with uncertainty-aware optimisation and other power system applications such as CO₂ emission modelling.

CRedit authorship contribution statement

Yogesh Pipada Sunil Kumar: Conceptualisation, Methodology, Software, Formal analysis, Investigation, Writing - original draft, Writing - review and editing. **S. Ali Pourmousavi:** Conceptualisation, Supervision, Project administration, Funding acquisition, Writing - review and editing. **Jon A.R. Liisberg:** Supervision, Writing - review and editing. **Julian Lesmos-Vinasco:** Supervision, Writing - review and editing.

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